CP115
Sesi 2026/2027
2 jam 30 minit

NAME: _____ **CLASS:** _____

KOLEJ MATRIKULASI KELANTAN
KEMENTERIAN PENDIDIKAN MALAYSIA

PRA PSPM 1
PROGRAM MATRIKULASI SISTEM DUA SEMESTER

COMPUTER PROGRAMMING
2 JAM 30 MINIT

JANGAN BUKA SOALAN INI SEHINGGA DIBERITAHU.
DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO.

ARAHAN KEPADA CALON: Kertas ini mengandungi 6 soalan. Jawab semua soalan di ruang yang disediakan.

INSTRUCTION TO CANDIDATE: This booklet consists of 6 questions. Answer all questions in the space provided.

No. Soalan	Markah
1	
2	
3	
4	
5	
6	
Jumlah	

Kertas soalan ini mengandungi **16** halaman bercetak, termasuk muka hadapan.
This booklet consists of **16** printed pages, including the cover page.

1. (a) Identify the most suitable steps in problem solving based on the given scenario

[3 marks]

Scenario	Problem Solving Step
An engineer creates flowcharts and algorithms to plan out the software logic before writing any code.	
A developer writes a user manual and technical documentation for the completed software.	
Lance needs to meet with his client to gather and document their requirements before developing the program.	

(b) Based on the following scenario, determine the type of errors.

[2 marks]

Scenario	Type of error
Trying to open and read a file that does not exist in the system	
The following code wants to check if the number is 1 num = input() #user enters 1 if num == 1: print(“The number is 1”)	

(c) Determine whether the following scenario is an input, process or output.

5 marks]

Scenario	Input, Process or Output
A user enters a student mark for calculating class average.	
The water pump is turned on when the water volume is low.	
Determining faulty or working electronic component	
A machine rejects unripe apples in the sorting machine	
Traffic light turns red	

(d) Write the input, process and output for the following scenario.

(i) Sweet Treats Bakery wants to calculate and display how much they earn if a loaf of bread costs RM6.00 per loaf with an additional RM1.50 per special filling.

[4 marks]

Input	
Process	
Output	

(ii) A system is developed for an agricultural industry. The client wants to know how much water and fertiliser is required for their crops based on the time of the day

Time	Water	Fertiliser
Day	2.0L	1.5L
Night	0.1L	0.2L

[6 marks]

Input	
Process	
Output	

(iii) A system is developed for an apple sorting facility. The system will count the number of ripe apples and unripe apples. Should there be a rotten apple, the system will stop to prevent contamination in the product.

[5 marks]

Input	
Process	
Output	

2. (a) Algorithms are the step-by-step process to solve a problem.

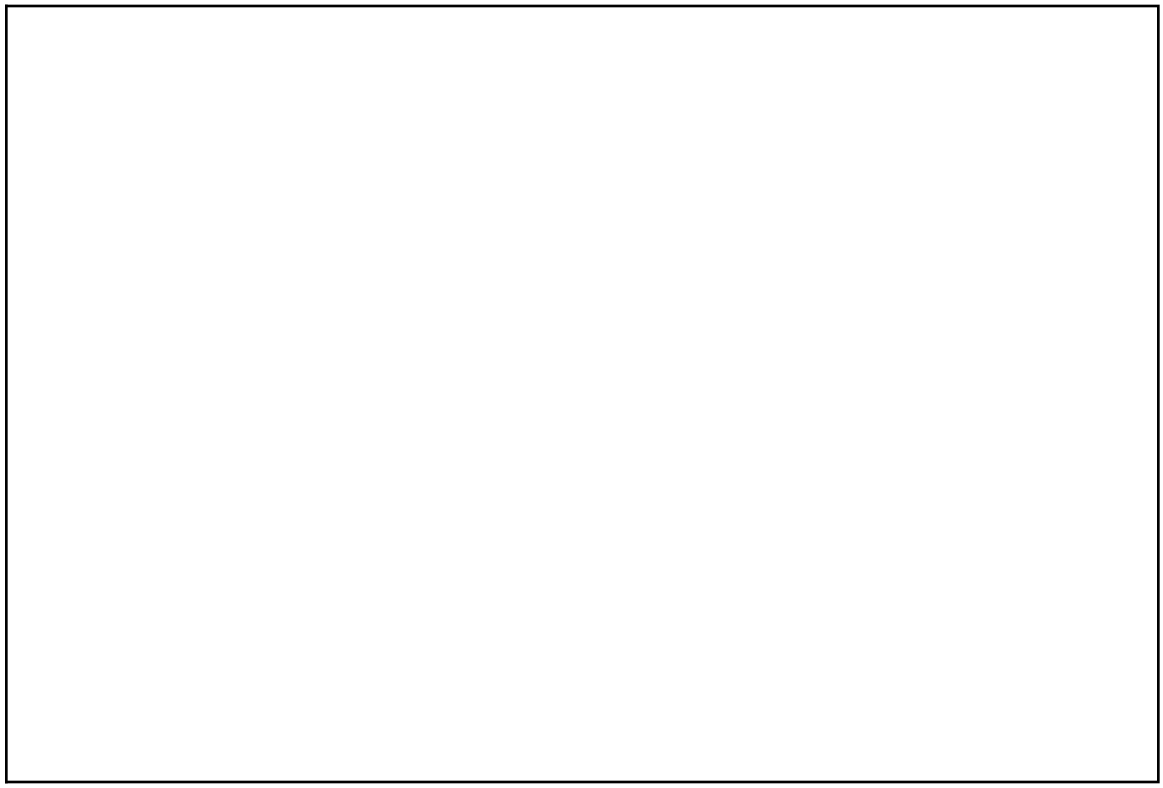
Describe one way of representing an algorithm.

[2 marks]

(b) Convert the following pseudocode in Figure 1 into flowchart

[5 marks]

```
Start
      x = 10
      While x > 0
            Read y
            If y > 0
                  z = x % y
            Else
                  z = y + x
            End if
            x = x - 1
      End while
      print z
Stop
```

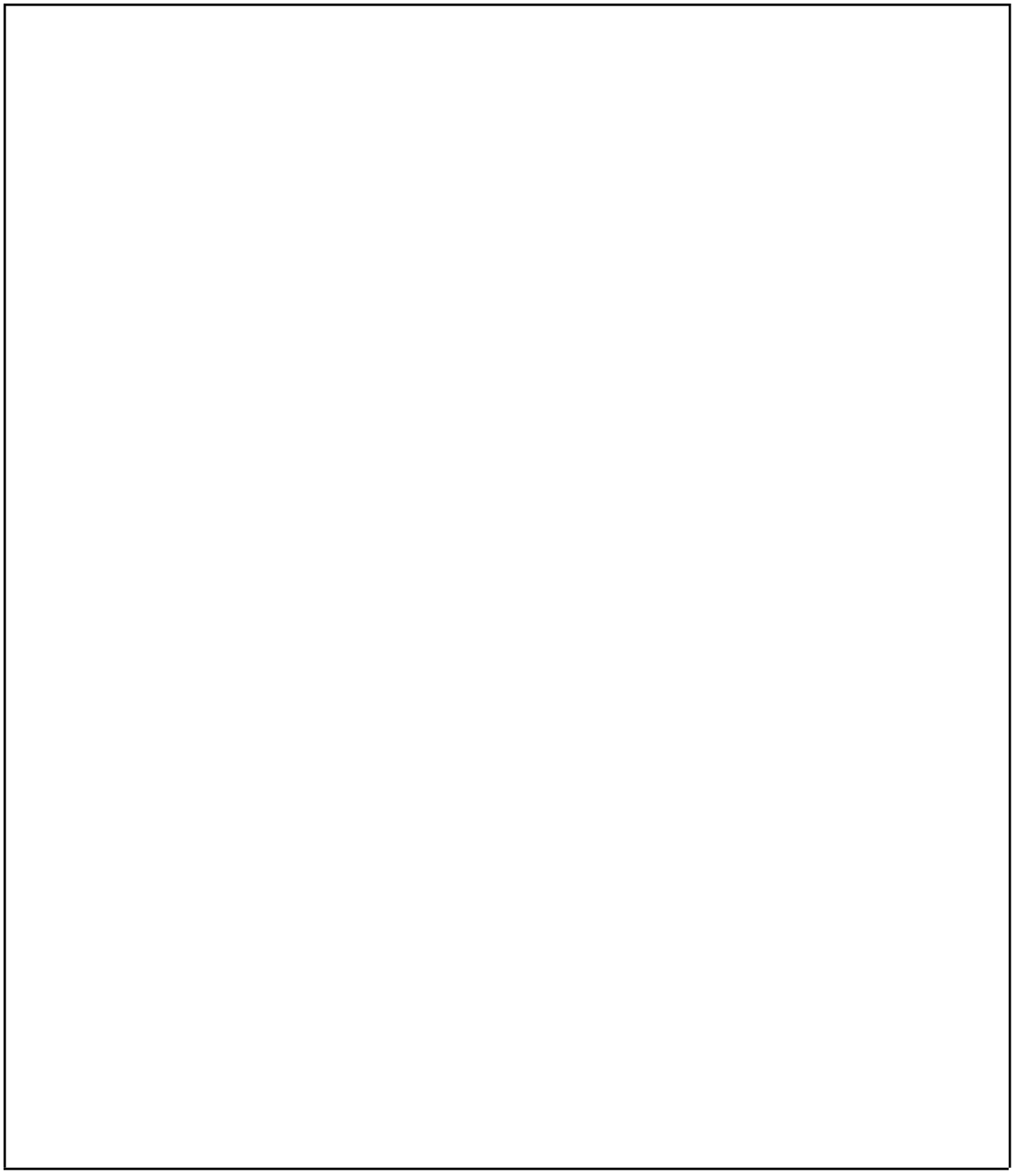


(c) A cinema calculates the ticket price based on the customer's age. The standard ticket price is RM15. Customers below 12 years old receive a 30% discount, while customers aged 12 and above pay the standard price. Write a pseudocode that prompts the user to enter their age, calculates the ticket price based on the age entered, and displays the final ticket price.

[6 marks]

(d) Create a flowchart where the user accepts the number of days and daily exercise minutes. The program will calculate and display the total accumulated exercise time based on the number of days entered and daily exercise minutes.

[5 marks]



3. (a) Define object oriented programming.

[2 marks]

(b) State an example of a compiled programming language.

[1 marks]

(c) Determine if the following identifiers are valid or not.

[3 marks]

Identifier	Valid/Invalid
print	
愛	
True	

(d) Write a suitable Python statement for the following statements.

[3 marks]

(i) $y = mx + c$

(ii) $d = \sqrt{(y_1 - y_2)^2 + (x_1 - x_2)^2}$

(iii) Write a suitable input statement to read a character from the user and store it in a variable named `grade`.

(e) The following code may contain error(s). Correct the code so the output will be
`Hello world`

```
A = "Hello"
B = "world"
print(A+B)
```

[2 marks]

4. (a) Write the output for the following code segment below if `num1 = 10`, `num2 = 2.0` and `num3 = 3`.

[2 marks]

```
a = num1 + num2 / num3
b = num3 ** num1 * num2
c = num1 + num3 // num2
d = (num1 % 5) != 0
print("Output 1:", a)
print("Output 2:", b)
print("Output 3:", c)
print("Output 4:", d)
```

(b) Describe repetition control structures.

[2 marks]

(c) Create a Python program that calculates and displays the area of a circle (π = 3.142)

[2 marks]

```
r = float(input("Enter radius: "))
```

(d) Complete the Python code segment according to the pseudocode given below.

[4 marks]

```
Start
    Read number
    sum = 0
    While number != 999
        if number % 2 == 0
            sum = sum + number
        end if
        Read number
    End while
    Print sum
Stop
```

5. (a) Create a program that calculates the shipping cost based on the following table. The user will enter the shipping category, type of transportation and the weight of the parcel and print out the total shipping cost at the end of the program.

[6 marks]

Category	Type of Transportation	Rate per 100gm
Normal	Sea	MYR 0.20
	Air	MYR 1.50
Expedited	Air	MYR 6.50

(b) Rewrite the following code segment using a `while` statement.

[2 marks]

.

```
for number in range(1,1001,2):
      print(number)
```

6. (a) Write a Python program that asks the user to enter a number. The program must repeat the input process up to 100 times. The program must calculate the sum of the current input and the previous input. If the sum is equal to 10, the program must stop the loop immediately.

```
Enter a number: 3
Enter a number: 4
Enter a number: 6
The sum of the current and previous numbers is 10.
The loop is terminated.
```

[6 marks]

(b) A game developer wants to develop a damage counter system to record the damage dealt by a character during a game. The program shall ask the user to enter the number of combos performed and the damage dealt for each combo. The program must calculate and display the **average damage**, **highest damage**, and **lowest damage** dealt by the character. Write a Python program to implement the damage counter system.

[12 marks]

```
Enter the number of combos: 4
Enter damage for combo 1: 120
Enter damage for combo 2: 250
Enter damage for combo 3: 180
Enter damage for combo 4: 90

--- Damage Counter Report ---
Average damage: 160.0
Highest damage: 250
Lowest damage: 90
```

END OF QUESTION PAPER